

T and F Distributions

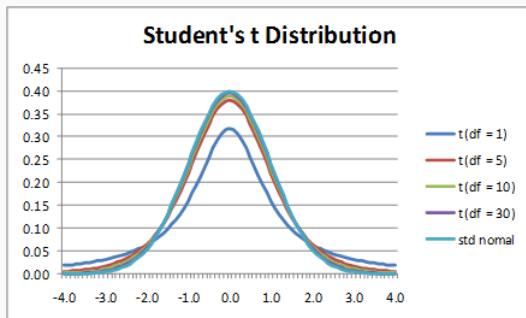
Student's t Distribution

- $W \sim N(0, 1)$, $V \sim \chi^2(r)$, $W \perp\!\!\!\perp V$,

$$T = \frac{W}{\sqrt{V/r}} \sim t_r.$$

- r : the number of degrees of freedom
- pdf:

$$g(t) = \frac{\Gamma(\frac{r+1}{2})}{\sqrt{\pi r} \Gamma(\frac{r}{2})} \left(1 + \frac{t^2}{r}\right)^{-\frac{r+1}{2}}, \quad t \in \mathbb{R}$$



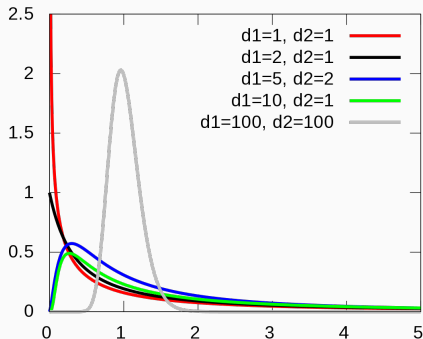
F Distribution

- $U \sim \chi^2(r_1)$, $V \sim \chi^2(r_2)$, $U \perp\!\!\!\perp V$,

$$W = \frac{U/r_1}{V/r_2} \sim F_{r_1, r_2}.$$

- (r_1, r_2) : First(second) degrees of freedom
- pdf:

$$g(w) = \frac{\Gamma(\frac{r_1+r_2}{2})\Gamma(\frac{r_1}{2})^{r_1/2}}{\Gamma(\frac{r_1}{2})\Gamma(\frac{r_2}{2})} \frac{w^{\frac{r_1}{2}-1}}{(1 + \frac{r_1}{r_2}w)^{\frac{r_1+r_2}{2}}}, \quad w > 0$$



What is the distribution of T^2 when $T \sim t_r$?

Student's Theorem

- Let $X_1, X_2, \dots, X_n$ be identically and independently distributed (i.i.d.) from $N(\mu, \sigma^2)$. Let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{sample mean}$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{sample variance}$$

Then,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$$

$$\bar{X} \perp\!\!\!\perp S^2$$

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$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2)$. Find c such that

$$\mathbb{P}\left(\frac{\bar{X}}{\sqrt{9S^2}} < c\right) = 0.95.$$